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Physics ATP Analysis By Vasumitra Gajbhiye

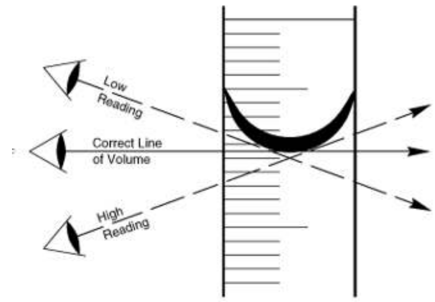
GENERAL

1. **Why take 5 readings and divide by 5 instead of 1 reading** → timing errors due to human reaction time have less effect./ reduce effect of errors in starting/stopping stopwatch.
2. **How to obtain more reliable readings** → repeat and average. In every seven or six marker do not forget to write this!
3. **When you want to suggest the relationship between two quantities and want to justify them** → within limits of experimental accuracy

Tip for seven/six markers → Do not forget to write plot a graph whenever possible and repeat and take average. Repeat for at least 6 values. Always write units when drawing the tables

7. **How to answer "explain how to use the readings to reach the conclusion" in seven/six markers** →
    - Compare readings in the table to see if change in factor(dependent variable) produces change in (Variable you are measuring, ex - speed, temperature)
    - plot line graph (with axes specified)
8. **How to accurately measure the volume of liquid in a measuring cylinder** →

- Bottom of the meniscus
- Line of sight perpendicular to the scale.



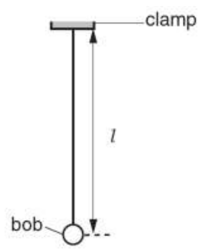
9. **How to measure diameter of a rod** →
    - Use of rod between two blocks and measure gap in at least 2 places and take average
    - Use micrometre , vernier callipers, measure several points and take average
10. **Precautions while measuring volume of wooden rod** → Wood might absorb water, measuring cylinder scale not precise
11. **How to calculate gradient of graph from a graph** → draw a triangle as large as possible of the graph. Then use the values and calculate the gradient
12. **Why it is useful to take a trial reading for a experiment** → to check if t(dependent variable) is measurable. to check if d(independent variable) value is appropriate. Establish range of d and t values.

LOAD & SCALE

1. **How to ensure if scale is horizontal** → measure distance of rule form bench at two ends and adjust until equal. Horizontal if equal
2. **It is sometimes difficult to position the load on the scale of the metre rule at the correct distance d from the pivot. Suggest one change to the apparatus to overcome this difficulty** → mark centre line of mass, suspend the load from loop of thread and then align with mark on rule
    - suspend load from loop of thread
    - (measure width of block and) add ½ width to 5.0 cm to find position for edge of block
    - mean value of marks at both edges of mass

- mark centre line of mass and align with mark on rule

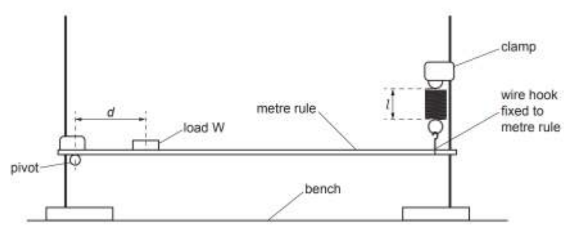
14. **How to make sure the length, l of the pendulum bob is as accurate as possible** →
    - Measure the top of the bob then add on half diameter measured with blocks and rule or callipers.
    - Measure the top and bottom of bob and average
    - View perpendicular to the scale
    - Rule parallel with string. Use of set-square
    - Keep the rule close to the pendulum
    - Measure from bottom of the clamp



4. **Explain briefly why the position of the pivot may not be exactly at the 50.0 cm mark of the rule** → center of mass of ruler not at the center

5. **What is the source of inaccuracy in this experiment** →

   test load not exactly 1.0 N / spring extension not linear / metre rule not uniform



TEMPERATURE

1. **How to get temperature reading from a thermometer as accurate as possible** →
    - Line of sight perpendicular to the thermometer scale. Wait until reading stop rising.
    - Stir before reading.
    - **IF container present then** - don't let thermometer touch the bottom/sides of the glass beaker/container
    - Wait for temperature to reach max before reading.
    - IF comparing the temperature of some liquid in two identical beakers - keep the thermometer at same depth
2. **What are the control variable while measure the rate of cooling/heating of water in a beaker** →
    - Same volume of water
    - Same initial temperature
    - Same surface area of container
    - Same diameter and height of container
    - Same room temperature
    - Same material of beaker
    - Same duration of experiment

16. **How to reduce loss of thermal energy from beaker** →
    - insulation
    - lid
    - lower starting temperature
    - higher room temperature
    - smaller volume of water
    - smaller surface area

17. **Experient for how lid affects the rate of cooling** → Repeat same experiment without a lid. calculate cooling rate and subtract cooling rate with lid.

RAY TRACE

1. **Possibility errors in ray trace experiments** →
    - Ray has finite thickness, rays not thin
    - Difficulty to align pins.
    - Pins too thick
    - Inaccuracies have more effect on smaller angles
2. **What are the precautions in ray trace experiments** →
    - Use thin pencil lines/ thin pins
    - View bases of pins
    - Large pin separation
    - Ensure pins are vertical/ upright/ erect

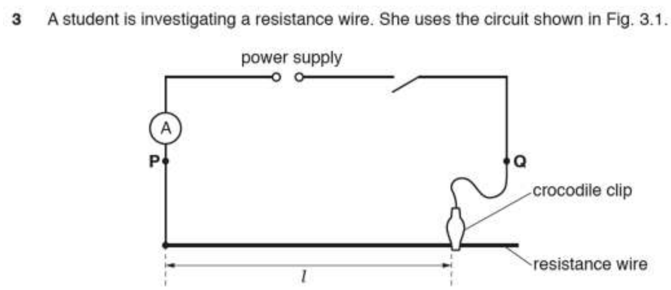
22. **Precautions for ray trace experiment** →
    - use thin pin
    - use thin lines OR sharp pencil
    - view bottom of pins OR keep pins upright
    - ensure pins far apart
23. **Why 2.7cm pin separation is not suitable** → pin separation should be as large as possible **for greater accuracy** / much larger / pin separation is too small /

ELECTRICITY

1. **What are the possible difficulties in resistance wire experiments** →

- Difficulties to judge position of crocodile clips

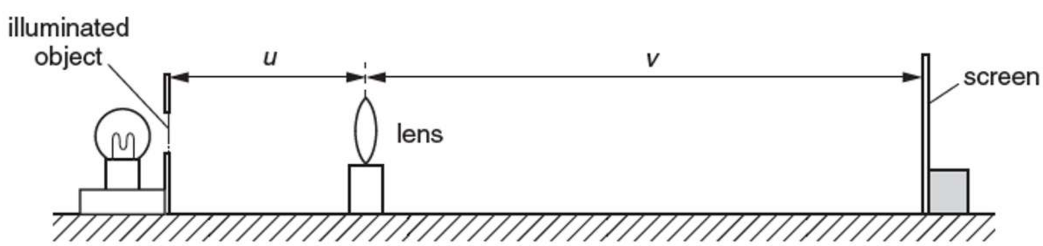
- Difficult to measure wire to nearest mm
- Contact between wire and crocodile clip not precise
- Difficult to interpolate readings on meters between marks



2. If u replace a resistance wire with a variable resistor, place the variable resistor in **series** with the circuit.
3. **What are the disadvantage of NOT using a variable resistor** →
    - cannot obtain continuous set of value.
    - less straightforward to change current
    - more difficult to obtain a greater number of values

LENS

15. **Precautions in lens experiments** →



- Darkened room/ brighter lamp/ no other lights
- Moving screen slowly to obtain best image
- Mark block to show centre of lens
- Place rule on bench/clamp rule
- Lens and object and screen vertical/perpendicular to the bench
- Repeat measurements/experiment and average